

# **Laser Stabilization for Quantum Computing**

***Sponsor:** University of Sydney Quantum Computing Group, Professor Mark Hempstead*

***Blast-Off Bronze:** Eric Rohloff, Matthew Dacey, Abe Nelson, Josh Wilkie*

## **1. Project Description**

Trapped-ion quantum computers heavily rely on the state of precise lasers to manipulate and read the state of atomic ions. These atoms encode information by behaving as qubits, which differ from bits in that they can be 0, 1, or a superposition of both states. Quantum experiments often run continuously for multiple days and the hardware needs stability. In order to achieve valuable experimental results, the wavelength of each laser must have precision of only a few nanometers over the entire experiment.

### **1.1 The Problem**

The primary challenge with trapped-ion quantum computers is the high sensitivity of the stabilizing lasers from drift and environmental effects. In order to correct drift, these systems use a method called Scanning Transfer Cavity Lock (STCL). Over time, the system changes the length of a cavity with mirrors at either end. When the length is an exact multiple of the laser's wavelength, the laser constructively interferes with itself and a higher voltage peak is read on a photodiode. When comparing the time differences to a stable reference laser the frequency drift can be determined. When drift occurs, a correction voltage is applied and the laser is shifted back to its desired wavelength. Current systems employ FPGAs or microcontrollers to run the STCL and provide feedback, but the algorithms used to control the lasers rely heavily on higher level programming languages such as C++ or Python. By using these higher level programming languages, there is a latency introduced that can be improved upon and greatly improve the speed in which quantum experiments can occur.

### **1.2 Project Requirements**

The main objective of Blast Off Bronze is to decrease the time in which the cavity scan and feedback loop occur in order to produce better laser stabilization. This can be achieved by migrating the laser control from software into hardware through the use of a FPGA. By implementing the laser control algorithms into Register Transfer Level (RTL) code on a FPGA, the system needs to reduce the latency from 4ms to 1ms or less. Additionally, the system must have a Python API in order for scientists with limited knowledge of RTL to interact with the system. Finally, existing solutions are able to stabilize up to 4 lasers, and we wish to match this with our solution. This project has been conducted with the support and collaboration of the

University of Sydney Quantum Science Group and Professor Mark Hempstead from the Tufts University Department of Electrical and Computer Engineering. They have supported Blast Off Bronze throughout this project by providing mentorship and guidance throughout the design process.

## **2. Design Choices**

### **2.1 Minimum Viable Product (MVP) and Customer Deliverables**

The Minimum Viable Product for the Blast Off Bronze is a functioning accelerated cavity scan stabilization loop. In order to meet the minimum specifications designated by the University of Sydney, the system must have peak-detection and drift correction implemented in RTL code on the FPGA. Additionally, all four configurable lasers and the reference laser must be implemented on the FPGA with control from the Python API. This process must all be done in under four milliseconds with a goal of one millisecond on latency. The final deliverables provided to the University of Sydney Quantum Science Group include a GitHub repository with extensive setup guides, Python and RTL code, and documentation of timing and physical hardware connections.

### **2.2 System Architecture**

In order to achieve the one millisecond of latency the feedback control loop was implemented into RTL code illustrated in the diagram below (Figure 1). The diagram depicts the path of a signal in the system. Light from the four configurable lasers and the stable reference laser enters an optical cavity and the length is swept over time. This results in peaks which are captured and processed in the FPGA. The processing of the incoming voltage peaks in RTL allows the FPGA to compare the difference between the reference and configurable lasers, sending a feedback signal back to the laser controller. Each controller then applies a voltage to control drift.

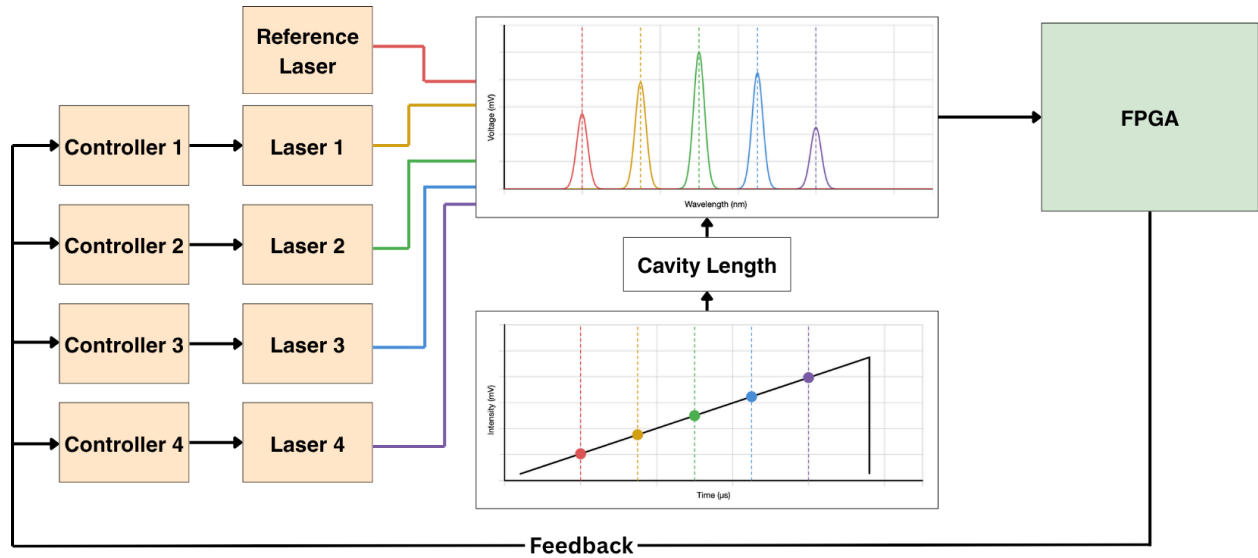


Figure 1 – Laser Stabilization Feedback Loop

### 2.3 Interface Design

The system is divided into two main categories: the higher-level system control, containing the API and inter-device communication, and the lower-level algorithm, containing the interface with the hardware. A host computer is used in order for researchers to interact with the system and control the lasers. The API was used to allow for a Python abstraction of the RTL code. These commands are sent over TCP to the ARM core on the FPGA. The ARM core can access shared memory on the FPGA through AXI registers. This allows the ARM Core to provide values to the Python API to display. The shared memory is also accessible by the DAC drivers, ADC driver, and hardware algorithms to allow for control of the feedback loop. These components interface with laser hardware, but, for this project, simulation data was used due to lack of access to physical laser hardware.

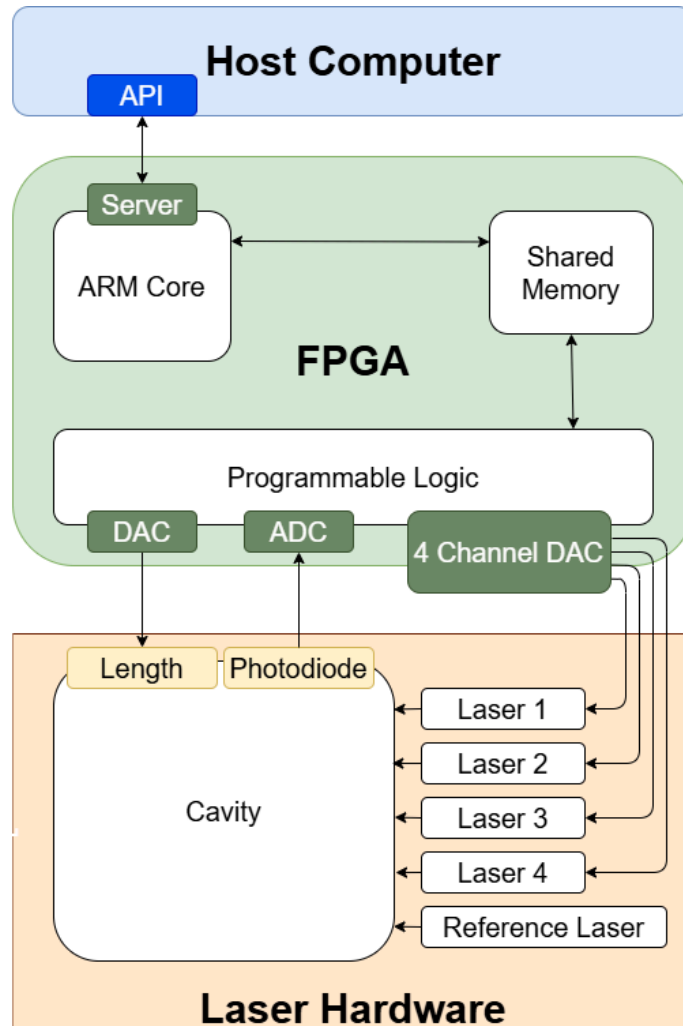


Figure 2 – System Interface Design

## 2.4 Technologies, Tools, and Methods

The required technology stack in order for this project to function was diverse. On the hardware side of the project we utilized the AMD AUP-ZU3 FPGA in order to receive and process data. This FPGA used System Verilog to define the architecture on the board in addition to this Xilinx Vivado was used for synthesis onto the FPGA fabric. A Function Generator and Oscilloscope was used in order for testing and sample data. For the software side of the project Python was primarily used in order to develop the user interface, API, and simulation. Jupyter Notebooks were also used in order to interact with the FPGA through the host computer. Finally, Git was used to have version control while working on the project in parallel.

Additionally, we spent time researching and purchasing the 3 analog peripherals needed to meet the timing requirements of this project. We ended up with 3 SPI-driven peripherals: a DAC and

ADC both clocked at 50 MHz and connected to the FPGA through PMOD connectors, and a 20MHz DAC connected to the GPIO pins of the FPGA using jumpers. Part numbers can be found in Section 3.

## **2.5 Acceptance Test Results**

The system underwent a series of tests to ensure the system met customer requirements and to ensure functionality. There were three main parts of the acceptance tests: FPGA and hardware, software, and system integration. The hardware testing involved making sure core functionality of the startup code and each component was properly working and could receive data. Software testing included simulation and running each command detailed in the API and making sure the results matched what was expected from the design. Finally for system integration, the TCP packet protocol and full system tests such as laser control and feedback were tested in order to ensure proper functionality.

## **3. Project Management**

### **3.1 Team Organization**

Executing the acceleration for the cavity scan required dividing up the group member tasks into smaller tasks in order to ensure the project was completed on time. This allowed for parallel completion of RTL code, Python development, and documentation. The responsibilities for each team member are listed below:

- Abe Nelson: RTL Code, Hardware Setup, Python Interface
- Matthew Dacey: RTL Code, Hardware Setup, Python Interface
- Eric Rohloff: Python Interface and Simulation, Documentation
- Josh Wilkie: Python Simulation and Design

### **3.2 Project Timeline**

The following table outlines the timeline of major project milestones during the process of the entire project.

| Date      | Project Milestone                        |
|-----------|------------------------------------------|
| September | Team Formation / Initial Sponsor Meeting |

|                    |                                                 |
|--------------------|-------------------------------------------------|
| October – November | Research and Functional Requirements Defined    |
| December           | Finalization of Hardware / Ordering Components  |
| Late January       | Hardware Setup and Initialization               |
| February – March   | Simulation Design and Cavity Scanning Algorithm |
| April              | Hardware and Algorithm Integration              |
| Early May          | Documentation and Handoff                       |

Figure 3 – Project Timeline

### 3.3 Budget Analysis

The development of the FPGA based laser stabilization system utilized resources purchased from our Senior Design budget. This budget encompassed \$500 provided by the Tufts Department of Electrical and Computer Engineering. These items are detailed in Figure 4 below.

| Component                  | Manufacturer      | Part Number      | Quantity | Cost         |
|----------------------------|-------------------|------------------|----------|--------------|
| <b>FPGA</b>                | RealDigital       | AUP-ZU3          | 1        | \$160        |
| <b>Laser Feedback DAC</b>  | Texas Instruments | DAC8564EVM       | 1        | \$50         |
| <b>Cavity Width DAC</b>    | Digilent          | PMOD DA3         | 1        | \$36         |
| <b>Laser Intensity ADC</b> | Analog Devices    | EVAL-AD7983-PMDZ | 1        | \$33         |
| <b>Total</b>               |                   |                  |          | <b>\$279</b> |

Figure 4 – Budget Allocation

### 4. Ethics and Social Impact

In order to evaluate the ethical implications of Laser Stabilization for Quantum Computing, we utilized the Awareness, Investigation, and Responding (AIR). This project focused solely on the laser stabilization and accelerating the time it takes to stabilize trapped-ion quantum computing systems, however, the end use of a quantum computing system carries many consequences on the global level.

## **4.1 Awareness**

Quantum computers have the potential to solve many problems through the use of complex modeling including medicine discovery and climate change modeling. On the other hand, quantum computing can pose a disastrous security and privacy threat. Quantum computing has the potential to run simulations that can solve widely used encryption algorithms. This means that billions of people can have sensitive data exposed to those who possess a quantum computer. Additionally, there is a large energy requirement for the laser systems required to use a trapped-ion quantum computer which may affect the environment.

## **4.2 Investigation**

Through our work on this project, we noticed that there is a large gap between those directly affected and indirectly affected by this system. The main impact this system has is on those in the University of Sydney Quantum Science Group. This project directly impacts the runtime and precision of their experiments, thus our system must be more precise and scalable than existing solutions in order to have an impact. Additionally our system must have proper checks to ensure the hardware used by the scientists is not damaged in any way. Those indirectly affected by our project involve the cybersecurity industry, regulatory bodies, and the public. Once quantum computing becomes more feasible, the need for regulation and safety guardrails increases as there may be large security vulnerabilities and data breaches from the use of a quantum computer by a bad actor.

## **4.3 Response**

In order to address these issues, Blast Off Bronze has conducted thorough research and testing into our solution to ensure our solution will have an impact on quantum computing research and not damage the sensitive equipment in the laboratory. All code has been simulated and documented to ensure that the lasers have not had any extreme voltage corrections which may damage the lasers. While our system helps boost quantum computing research productivity, the broader issues that may occur with quantum computing are out of our hands. As students and engineers, we must be engaged and committed with any efforts to regulate quantum computing and enhance security of modern database systems.



## **5. Learning Reflection**

### **5.1 Teamwork and Collaboration**

At the beginning of the Senior Design Project, our group exclusively worked together on tasks for the project. This was required, as it took a lot of collaboration and research to derive the project requirements. Additionally, our group was blocked on trying to purchase the correct hardware. This meant that any technical work, such as algorithm development or driver design, was difficult to complete without all of the group having input. Once the scope of the project widened when we purchased the hardware, our group restructured how we approached the work, moving towards a more parallel approach. This allowed us each to specialize on different aspects of the project: simulation, Python API, and RTL. The transition to parallel teams also allowed for our group to be more productive in developing the final system.

### **5.2 Main Takeaways**

The most significant learning and takeaway from this experience was understanding the planning and management required to execute a large scale project. Translating the complicated system requirements from our sponsor into hardware specifications proved more challenging than initially thought. We learned that programming and design was only one part of the project and organizing the integration of each piece required additional planning and consideration. Time management and communication also required a large amount of planning to ensure that our group was on the same page, our sponsors were in the know, and technical documents were submitted to the right place and on time.

### **5.3 Next Steps**

The hardware accelerated feedback logic has been successfully implemented using sample data provided by the University of Sydney. Future work for this project will involve ensuring the system can perform feedback on a system with actual laser hardware. Integrating the FPGA with live lasers and photodiodes in the laboratory may produce bugs or logical issues that may need some tweaking. Enhancing the RTL code to handle real world noise and drift that the sample data cannot replicate may also need to be considered. These next steps are crucial to perfecting this system and enhancing quantum computing research.

## References

[1] S. Subhankar, A. Restelli, Y. Wang, S. L. Rolston, and J. V. Porto, “Microcontroller based scanning transfer cavity lock for long-term laser frequency stabilization,” 2018, doi: [10.48550/ARXIV.1810.07256](https://doi.org/10.48550/ARXIV.1810.07256).

[2] E. Pultinevicius et al., “A scalable scanning transfer cavity laser stabilization scheme based on the Red Pitaya STEMLab platform,” Review of Scientific Instruments, vol. 94, no. 10, p. 103004, Oct. 2023, doi: [10.1063/5.0169021](https://doi.org/10.1063/5.0169021).

## Appendices

Github Repository (<https://github.com/ericrohloff/EE98-LaserStabilization.git>)